

Experiment 4: Step-Up DC-DC Converter

Part 2: Closed-Loop Cascaded Boost DC-DC Converter

ECEN 5517-012

Khalid Shahba

Stewart Rojec

BENCH 5

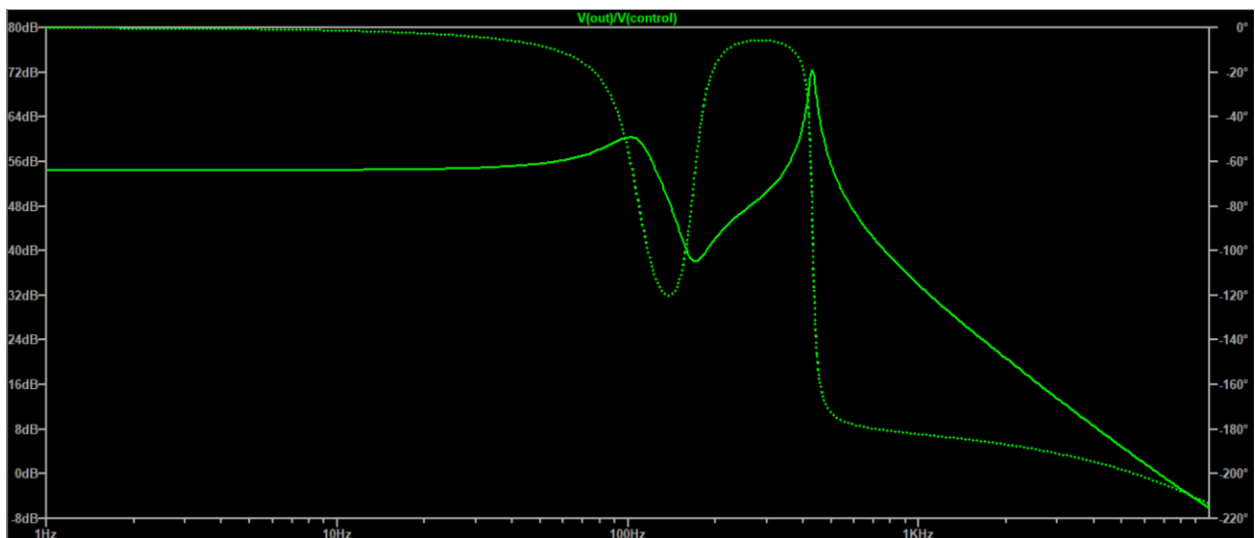
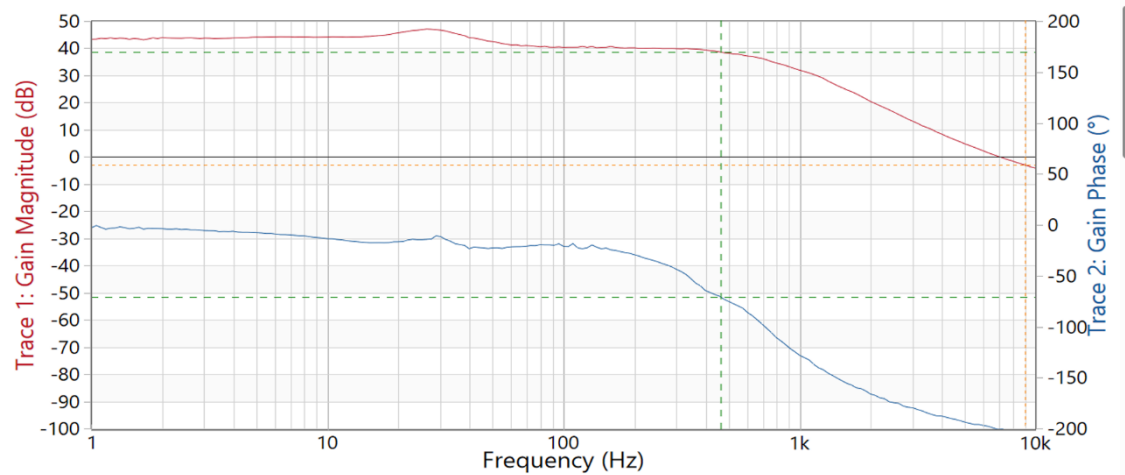
I.Introduction

The purpose of this experiment is to implement closed-loop regulation to the cascaded boost converter from Part 1. To accomplish this, we'll add a voltage divider and integral compensator to the 150 V output and feed it back to the PWM driver input based on the circuit we designed in the prelab.

II.Experimental Procedure and Results

a. Measurement of Control-to-Output Transfer Function

i. Measured Bode Plots



Comparing To Simulation at Selected Frequencies

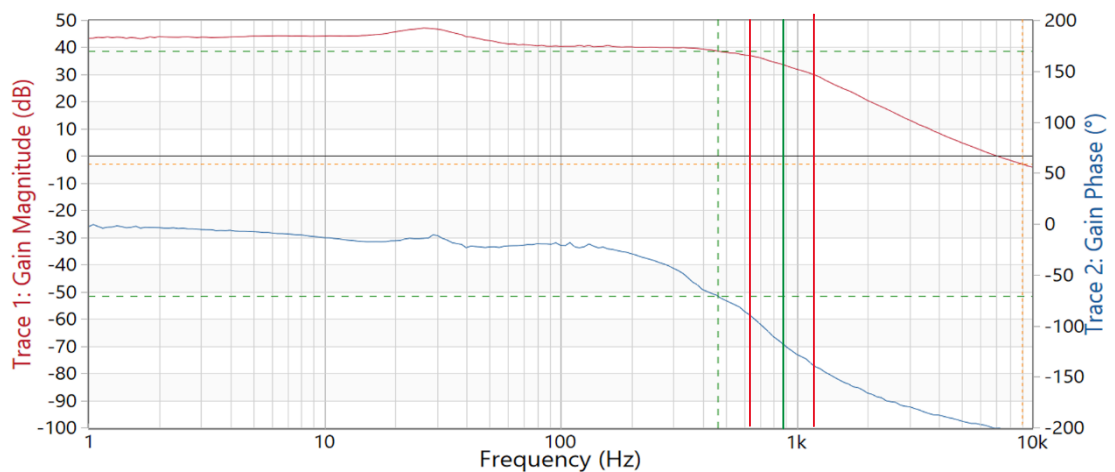
Frequency	Gain (dB) [Simulation]	Phase (deg) [Simulation]	Gain (dB) [Measured]	Phase (deg) [Measured]
10 Hz	54.47	-1.25	44.29	-13.24
100 Hz	60.3	-52	40.46	-20.78
464.159 Hz	61.78	-163.61	38.719	-70.643
1000 Hz	33.8	-182	32.1	-127.69
10000 Hz	-6	-213.5	-3.98	-205.62

The open loop Bode plots only seem to match in the left and right asymptote behavior and have a big discrepancy in the behavior between 100 and 1000 Hz. The phase also appears to be lagging in the measured plot. This is most likely due to how the Bode 100

Analyzer attenuates signals above 100 Hz, which would filter out the complex poles and zero that we see in the simulated plot. That attenuation could also explain the phase discrepancy, depending on how the Bode 100 accomplishes it.

b. Analysis of Control to Output Transfer Function

We can see that the measured Bode plot appears to have a single complex pole at around 900 Hz. To find the properties of the straight-line approximation, we estimate the bandwidth of the second order pole using the -3 dB frequency and the frequency where the slope is -40 dB / decade, shown below.



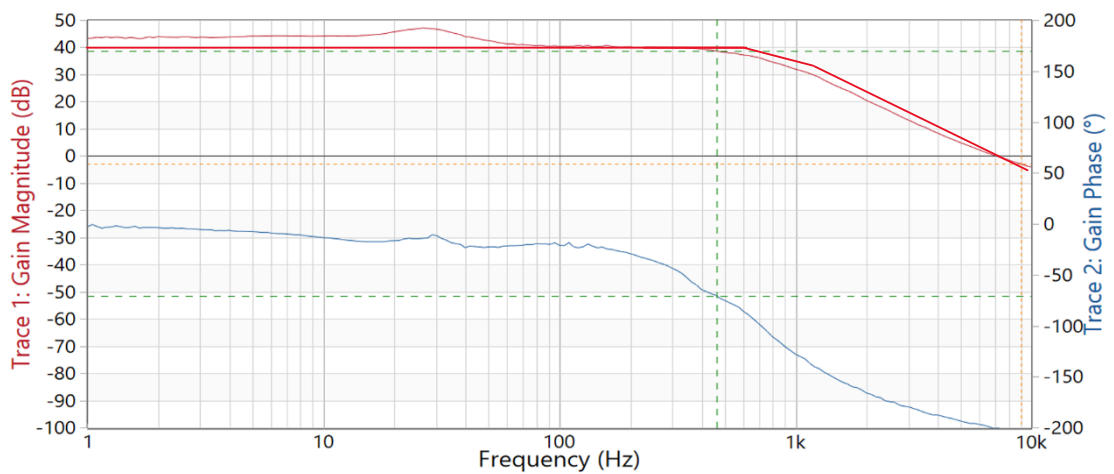
From the graph and imported data from this graph, we can see the corner frequencies of the overdamped pole. The ω_0 and Q factor are calculated below.

$$\omega_{c1} = 630 * 2\pi = 3958.4 \quad \omega_{c2} = 1200 * 2\pi = 7539.8$$

$$G_{vc} = \frac{K}{(s+3958.4)(s+7539.8)} = \frac{K}{s^2+11498s+29845544.32} \rightarrow \omega_0 = 869.48 \text{ Hz}, Q = 0.475 = -6.46 \text{ dB}$$

The gain before the asymptote is 40 dB, so we can conclude that the gain is 100, meaning our final transfer function is as follows:

$$G_{vc} = \frac{100}{\left(\frac{s}{3958.4} + 1\right)\left(\frac{s}{7539.8} + 1\right)}$$



Here, we begin the analytical solution and equivalent circuit model of the cascaded boost. First it begins with a completely general tabulation of inductor voltages and capacitor currents during each part of the switching cycle:

$$\begin{aligned}
 A &= V_{L1} = V_I(t) - I_{L1}(t)R_{L1} - I_{L1}(t)R_{on1} \\
 B &= V_{L2} = V_K(t) - I_{L2}(t)R_{L2} - I_{L2}(t)R_{on2} \\
 C &= i_{C1} = -I_{L2}(t) \\
 D &= i_{C2} = -\frac{V_O(t)}{R_L} \\
 E &= V_{L1} = V_I(t) - I_{L1}(t)R_{L1} - V_{D1} - V_K(t) \\
 F &= V_{L2} = V_K(t) - I_{L2}(t)R_{L2} - V_{D2} - V_O(t) \\
 G &= i_{C1} = I_{L1}(t) - I_{L2}(t) \\
 H &= i_{C2} = I_{L2}(t) - \frac{V_O(t)}{R_L} \\
 &\dots
 \end{aligned}$$

Next, we add an understanding that the average inductor voltage and average capacitor current in a switching cycle are like digital samples of voltages and currents. For switching frequencies much higher than the frequencies being tested here, we can assume it to be a continuous time-averaged signal. Now, we define these time-averages signals as having a “DC bias”/operating point with a small signal at frequency much lower than the switching frequency. Each of these period-averaged signals in time has a duty cycle $d(t)$ that occurred in it that help create the DC and AC equations for this model. Since the derivatives of the inductor current and capacitor voltage of the DC equations are 0, we get the DC operating point equations, which for a lossy case look like this:

$$\begin{aligned}
 eqnL1dc &= 0 = V_I - V_{D1} - V_{C1} + D V_{C1} + D V_{D1} - I_{L1} R_{L1} - D I_{L1} R_{on1} \\
 eqnL2dc &= 0 = V_{C1} - V_{C2} - V_{D2} + D V_{C2} + D V_{D2} - I_{L2} R_{L2} - D I_{L2} R_{on2} \\
 eqnC1dc &= 0 = I_{L1} - I_{L2} - D I_{L1} \\
 0 &= I_{L2} - \frac{V_{C2}}{R_L} - D I_{L2}
 \end{aligned}$$

For $V_I = 12$, $R_L = 500$, and all lossy elements taken into account, and for solving $V_{C2} = 150$, the following DC operating point values are established that are used in creating the AC equivalent circuit model.

VC2	D	IL1	IL2	VC1
150	0.7229	3.9069	1.0826	42.2029

Now, the AC equations are as follows for the 4 reactive elements.

For the inductors $L di/dt$ is

For inductor 1:

$$42.9486 d - 0.0214 i_{l1} - 0.2771 v_{c1} + v_i$$

For the capacitors, $C dv/dt$ is

For capacitor 1:

$$0.2771 i_{l1} - 3.9069 d - i_{l2}$$

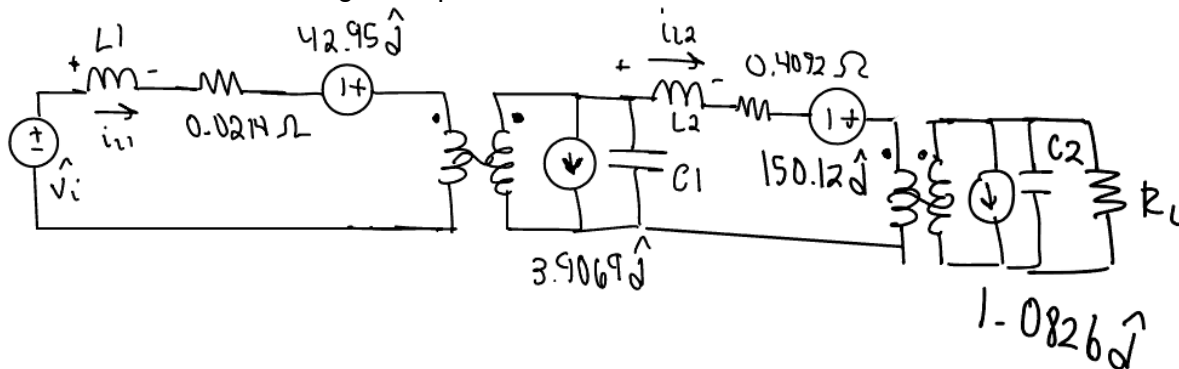
For Inductor 2:

$$150.1262 d - 0.4092 i_{l2} + v_{c1} - 0.2771 v_{c2}$$

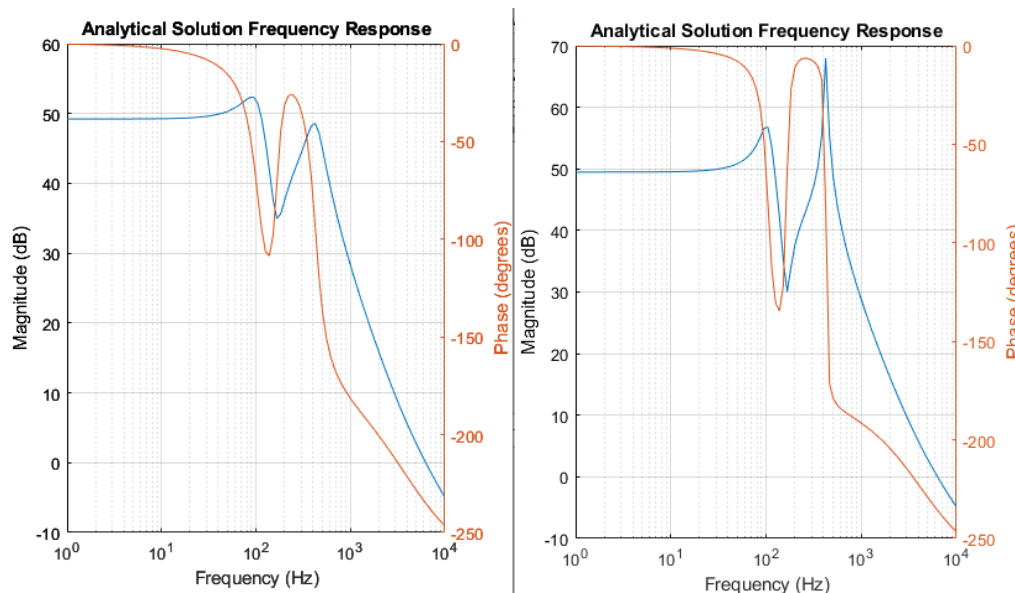
For capacitor 2:

$$0.2771 i_{l2} - 1.0826 d - 0.0020 v_{c2}$$

This results in the following AC equivalent circuit model:



This circuit model was then analyzed using superposition with the first attempt yielding the following AC response for $G_{vd}(s)$ when all losses are incorporated (left) and the AC response when only inductor series resistances were included to more closely resemble the model in LTSpice:



It is clear that the added losses in the more complete solution remove some of the Q in the magnitude response. These solutions match the LTSpice simulation quite well except for the fact that this is the duty cycle to output voltage gain, not control to output gain. That is quite straightforward to add. The control to duty cycle gain is a flat -6 dB attenuation added to these plots. For some reason, this results in a LF gain of only 43.5 dB where the LTSpice simulation had 54.4 dB. There is a factor of 3.5 of gain less in these models than in LTSpice models but these ones better resemble the real-world LF gain anyway.

Incorporating all the real world loss, the $G_{vd}(s)$ is

$$\frac{3.6893e+18 (-6.1906e+44 s^3 + 1.5845e+49 s^2 + 3.9881e+51 s + 1.7382e+55)}{6.9620e+58 s^4 + 9.9018e+61 s^3 + 5.5646e+65 s^2 + 2.0652e+68 s + 2.2178e+71}$$

, or divide by 2 to get

$G_{vc}(s)$

Incorporating only inductor losses, the $G_{vd}(s)$ is

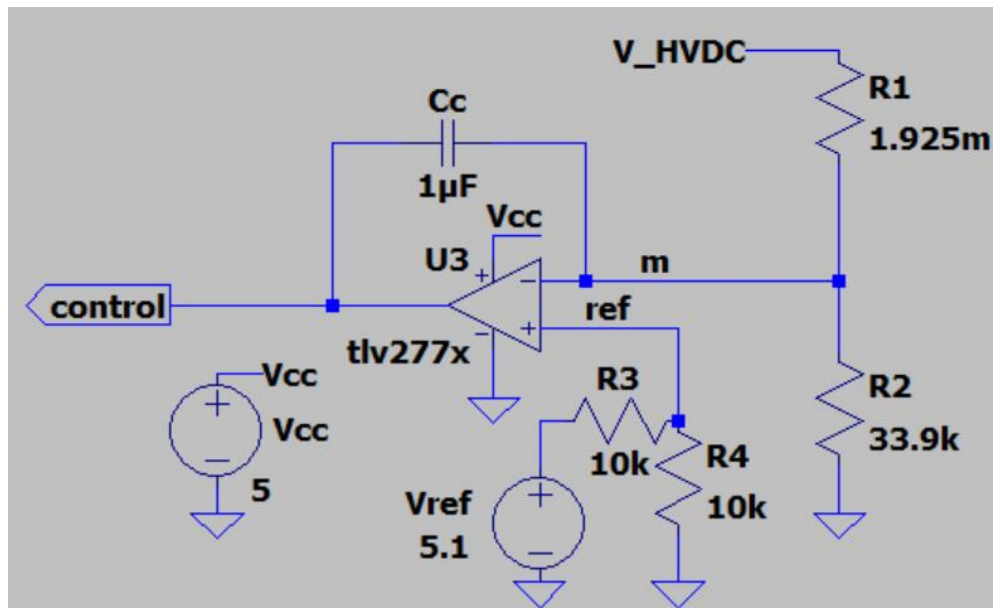
$$\frac{1.2627e+28 (-2.7135e+146 s^9 + 7.1278e+150 s^8 + 1.4434e+152 s^7 + 1.0497e+158 s^6 + 2.9678e+160 s^5 + 4.5400e+164 s^4 + 1.2237e+167 s^3 + 5.1301e+170 s^2 + 7.8558e+172 s + 1.4356e+176)}{1.0444e+170 s^{10} + 7.5961e+172 s^9 + 2.2203e+177 s^8 + 1.3327e+180 s^7 + 1.6281e+184 s^6 + 7.6974e+186 s^5 + 4.5405e+190 s^4 + 1.5434e+193 s^3 + 3.1699e+196 s^2 + 5.3939e+198 s + 6.0848e+201}$$

, again divide by 2 to get $G_{vc}(s)$.

What is bizarre is that by eliminating some sources of loss, the response becomes more accurate to that of the spice model as expected, but it becomes a 10th order transfer function.

c. Compensator Circuit Design

The compensator we initially designed met the phase margin requirements in simulation. However, we did need to redesign it to solve our regulation problems. We increased R1 several times to decrease our crossover frequency and increase the phase margin in the hopes it would move to a range that successfully regulates at 150 V. We eventually landed on a value 1.925 MΩ, and experimented with R2 values until we got as close as possible to 150 at the output, which ended up being 33.9 kΩ. The diagram is shown below.

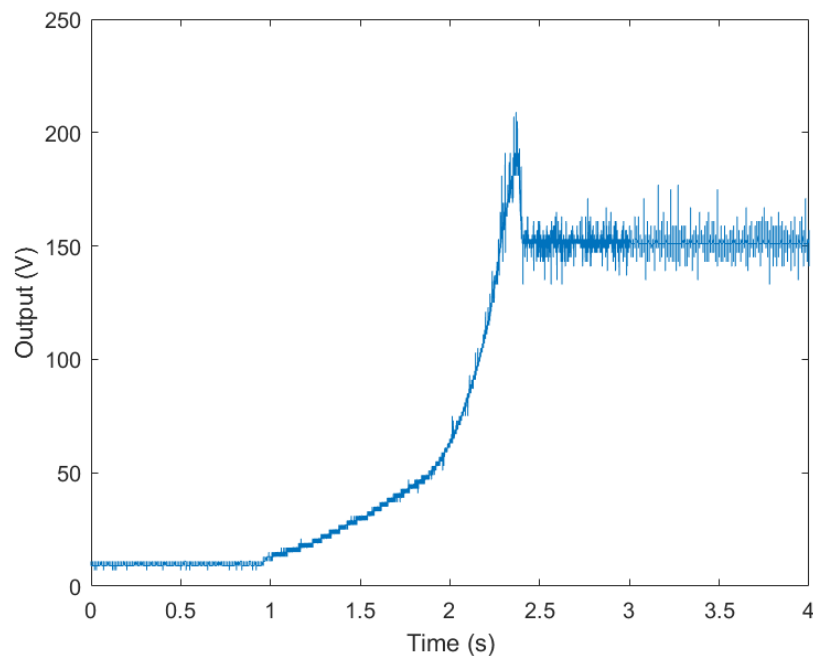


d. Start Up of the Closed-Loop Regulated Converter

The closed-loop test required several iterations and rounds of troubleshooting before it operated properly. We needed to increase our maximum duty cycle several times, since the converter was failing to regulate, even at an incorrect output voltage, at any input less than ~15 V. We also needed to change the circuit to reduce noise on the op-amp output, which was disrupting the 3525 chip's ability to accurately modulate duty cycle. The values that eventually worked gave us a duty cycle of 81.07 % and 87.4 kHz, as shown below.



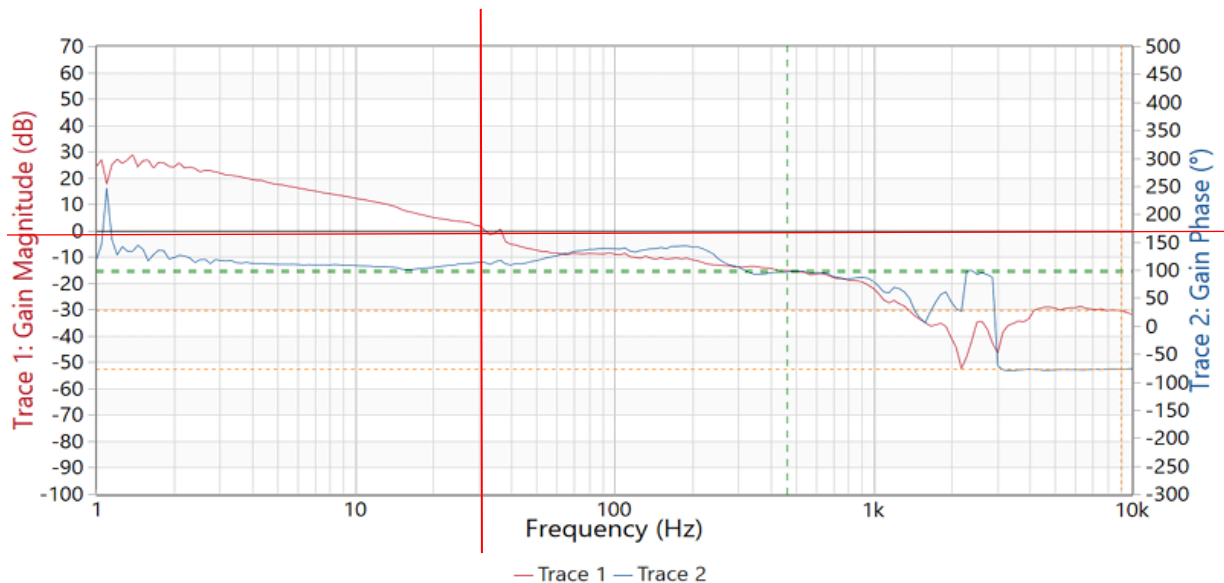
We also implemented a soft-start circuit on the 3525, using a larger-than-typical capacitor of 33 μF as recommended by Aarranon when trying to troubleshoot our maximum duty cycle troubles, during which the power supply was prone to maxing out to constant current mode. The measured soft-starting output is shown below.



The soft-start time is a little longer than specified, **1.387 seconds** instead of 1, but it was necessary for our circuit to give a longer amount of time for the voltage to ramp up to not max out the power supply's current rating.

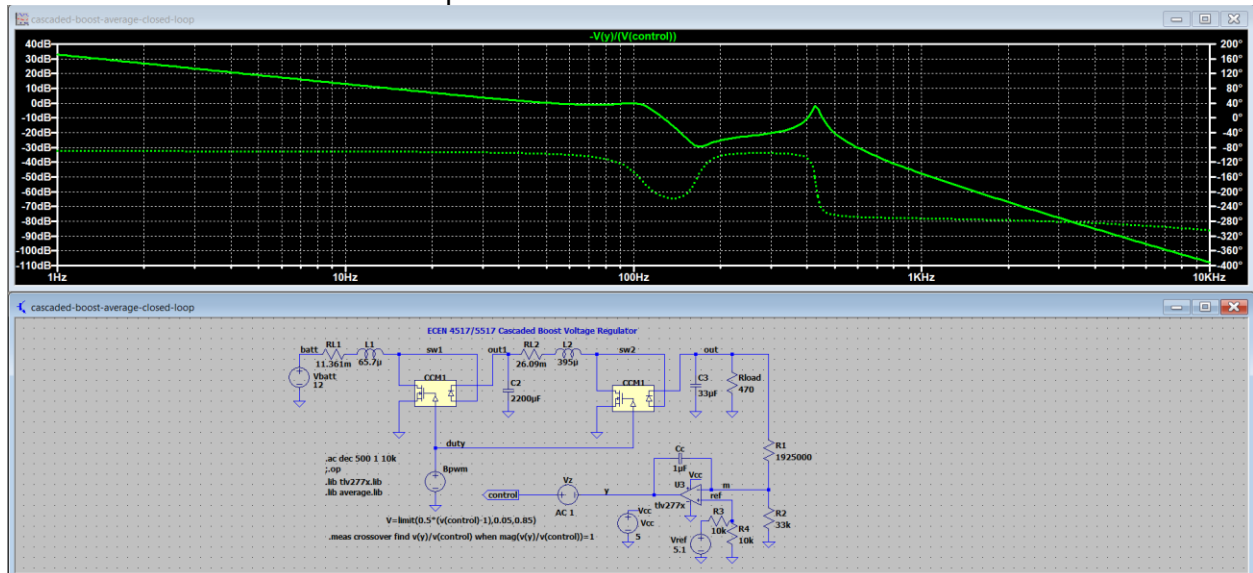
e. Measuring Loop Gain

i. Measured Loop Gain Bode Plots



Cross Over Frequency	Phase Margin
31 Hz	$180 + 112.5 = 292.5$ degrees

ii. LTSpice Simulation



Cross Over Frequency	Phase Margin
50.2 Hz	$180 - 97.4 = 82.6$ degrees

iii. Comments

Weirdly, the phase margin is a whopping 292.5 degrees. It must just be that we connected the output of the isolation transformer in reverse so that when the signal came back it had come in 180 degrees out of phase more than expected. **As a result, our probable actual phase margin is $180 - 112.5 = 67.5$ degrees.** This is below the minimum spec of 70 degrees but since the

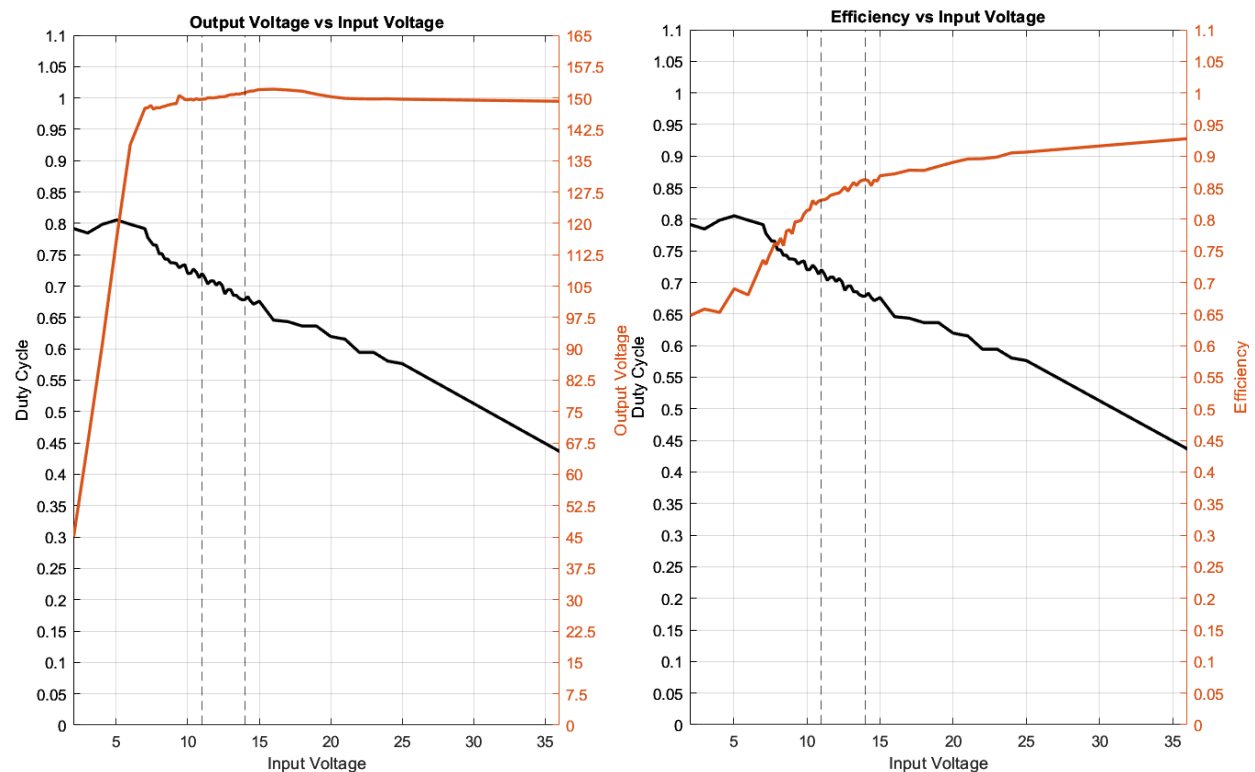
converter just works so well, we decided to move on. Also, please note, the schematic here has 33k as R2 as that is the resistor value needed to achieve the desired output voltage of 150V in steady state of the converter. In reality, it was changed to 33.9k in order to more accurately output 150V in the real world. The resistor divider was initially designed for a cross-over frequency of 78 Hz where the phase margin of the loop gain would have been such that the phase margin was exactly 70 degrees. Following advice of the TAs, we gave ourselves a margin of error with the selection of R1 and we clearly ended up needing it as our measured phase margin appeared to just be below spec still.

f. Regulation Performance

- i. 45W Output Voltage Regulation vs Input Voltage (our load was not exactly 500Ω) to get exactly 45W out.

Input Voltage	Input Current	Output Voltage	Output Current	Deviation from 150V (%)	Duty Cycle	Efficiency	Input Power	Output Power	Intermittent FS Check
2.03	2.75	45.2	0.08	69.8667	0.79167	0.647738468	5.5825	3.616	
3	4.1	67.21	0.1204	55.1933	0.7847	0.657893008	12.3	8.092084	FS = 86.8k
4.04	5.65	91.4	0.163	39.0667	0.7986	0.652685534	22.826	14.8982	
5.04	7.16	116.7	0.2134	22.2000	0.8055	0.69011539	36.0864	24.90378	
5.98	8.59	138.83	0.2517	7.4467	0.7986	0.680255703	51.3682	34.943511	
7.01	7.9	147.55	0.276	1.6333	0.7916	0.735365391	55.379	40.7238	
7.2	7.6	147.66	0.27	1.5600	0.7777	0.728585526	54.72	39.8682	
7.42	7.32	148.2	0.272	1.2000	0.7708	0.742167823	54.3144	40.3104	
7.6	6.97	147.3	0.27	1.8000	0.7655	0.750792872	52.972	39.771	
7.8	6.71	147.7	0.27	1.5333	0.7655	0.761951164	52.338	39.879	
8	6.52	147.6	0.2683	1.6000	0.7517	0.75922316	52.16	39.60108	
8.2	6.32	147.87	0.27	1.4200	0.7517	0.770394026	51.824	39.9249	
8.4	6.16	148.02	0.265	1.3200	0.7431	0.758064703	51.744	39.2253	
8.6	5.96	148.28	0.27	1.1467	0.7431	0.781090994	51.256	40.0356	
8.8	5.79	148.42	0.269	1.0533	0.7372	0.783580232	50.952	39.92498	
9	5.63	148.6	0.265	0.9333	0.7372	0.777165976	50.67	39.379	
9.2	5.48	148.64	0.27	0.9067	0.7361	0.796033005	50.416	40.1328	
9.4	5.47	150.6	0.272	0.4000	0.7292	0.796670427	51.418	40.9632	
9.6	5.29	150.15	0.27	0.1000	0.7324	0.798292769	50.784	40.5405	FS = 88.0K
9.8	5.12	149.63	0.271	0.2467	0.7343	0.808149912	50.176	40.54973	
10	4.98	149.55	0.271	0.3000	0.7203	0.813816265	49.8	40.52805	
10.2	4.86	149.7	0.27	0.2000	0.7203	0.815359477	49.572	40.419	
10.4	4.73	149.5	0.273	0.3333	0.7273	0.82967759	49.192	40.8135	
10.6	4.65	149.82	0.271	0.1200	0.7222	0.823721242	49.29	40.60122	
10.8	4.53	149.6	0.271	0.2667	0.7133	0.828664868	48.924	40.5416	
11	4.44	149.7	0.271	0.2000	0.7203	0.830644963	48.84	40.5687	
11.2	4.36	149.73	0.271	0.1800	0.7133	0.830947534	48.832	40.57683	
11.4	4.28	150.03	0.271	0.0200	0.7034	0.833295007	48.792	40.65813	FS = 86.2k
11.6	4.2	150.04	0.272	0.0267	0.7083	0.837661741	48.72	40.81088	
11.8	4.12	150.01	0.272	0.0067	0.7083	0.839285832	48.616	40.80272	
12	4.05	150.1	0.2722	0.0667	0.7014	0.840683539	48.6	40.85722	
12.2	3.98	150.25	0.272	0.1667	0.7063	0.841667353	48.556	40.868	
12.4	3.91	150.27	0.273	0.1800	0.7014	0.846128826	48.484	41.02371	
12.6	3.84	150.33	0.274	0.2200	0.6875	0.851323165	48.384	41.19042	

12.8	3.79	150.6	0.272	0.4000	0.6944	0.84439314	48.512	40.9632	
13	3.73	150.78	0.274	0.5200	0.6944	0.852004949	48.49	41.31372	
13.2	3.66	150.8	0.275	0.5333	0.6853	0.858378871	48.312	41.47	
13.4	3.63	150.94	0.275	0.6267	0.6853	0.853346902	48.642	41.5085	FS = 87.4k
13.6	3.55	150.92	0.275	0.6133	0.6806	0.859631317	48.28	41.503	
13.8	3.5	151.1	0.2755	0.7333	0.6783	0.861864389	48.3	41.62805	
14	3.45	151.22	0.2755	0.8133	0.6784	0.862548861	48.3	41.66111	
14.2	3.42	151.5	0.276	1.0000	0.6831	0.861008154	48.564	41.814	FS = 88.03k
14.4	3.37	151.66	0.273	1.1067	0.6761	0.853181256	48.528	41.40318	
14.6	3.32	151.66	0.2756	1.1067	0.6713	0.862301865	48.472	41.797496	
14.8	3.28	151.88	0.275	1.2533	0.6736	0.860394693	48.544	41.767	
15	3.23	152	0.277	1.3333	0.6761	0.869019608	48.45	42.104	
16	3.02	152.1	0.277	1.4000	0.6458	0.871930877	48.32	42.1317	
17	2.83	151.9	0.278	1.2667	0.6434	0.877742673	48.11	42.2282	
18	2.65	151.62	0.276	1.0800	0.6364	0.877298113	47.7	41.84712	
19	2.48	150.9	0.276	0.6000	0.6364	0.883879457	47.12	41.6484	FS = 87.4k
20	2.33	150.3	0.276	0.2000	0.6197	0.890188841	46.6	41.4828	
21	2.2	149.9	0.276	0.0667	0.6154	0.895506494	46.2	41.3724	
22	2.09	149.8	0.275	0.1333	0.5944	0.895933014	45.98	41.195	
23	2	149.76	0.276	0.1600	0.5944	0.89856	46	41.33376	FS = 87.4k
24	1.91	149.8	0.277	0.1333	0.5804	0.905205061	45.84	41.4946	
25	1.83	149.7	0.277	0.2000	0.5764	0.906380328	45.75	41.4669	
36.23	1.25	149.2	0.2817	0.5333	0.4336	0.92806271	45.2875	42.02964	



Over the desired qualification range of $V_{in} = 11V$ to $14V$, the converter had a minimum and maximum output voltage of $149.7V$ to $151.5V$ (-0.2% to 1.0%). Below roughly $8V$ in, we reached our peak duty cycle of the PWM IC and were not able to adjust PWM to reach $150V$. The op-amp in the compensator summarily railed and resulted in the input to output voltage conversion ratio being a

constant $\frac{1}{(1-D_{max})^2}$, where for a $D_{max} = 81.07\%$, the ratio was 27.9. For 2V in, we saw 45.2V out, which is only a conversion ratio of 22.6, but this real-world conversion ratio has losses and the other assumes ideality. This can be seen by the straight line from 0 to 8V on the left plot. **Our efficiency in this qualification range was from 83.06% to 86.25%**, getting larger as less current was required at the input for the same power at the output. This trend continued at our extremes as well.

We have thus established **a minimum input voltage** required for this cascaded boost converter such that at least 149V is at the output – **roughly 9.2 to 9.4V**. This is where the maximum PWM IC duty cycle begins to fall as compensation occurs (functioning feedback in integral compensator op-amp). **The maximum input voltage could hypothetically be up to 63V** as the 22 mF capacitor is the voltage-rating limiting component on the input. Since our supply could only go up to 36.23V, we just tested up to there and **got an efficiency of 92.8% when 42W were dissipated across the load at 149.2V out**. It can be seen for any input voltage above 9.2 to 9.4, regulation occurs to spec as well.

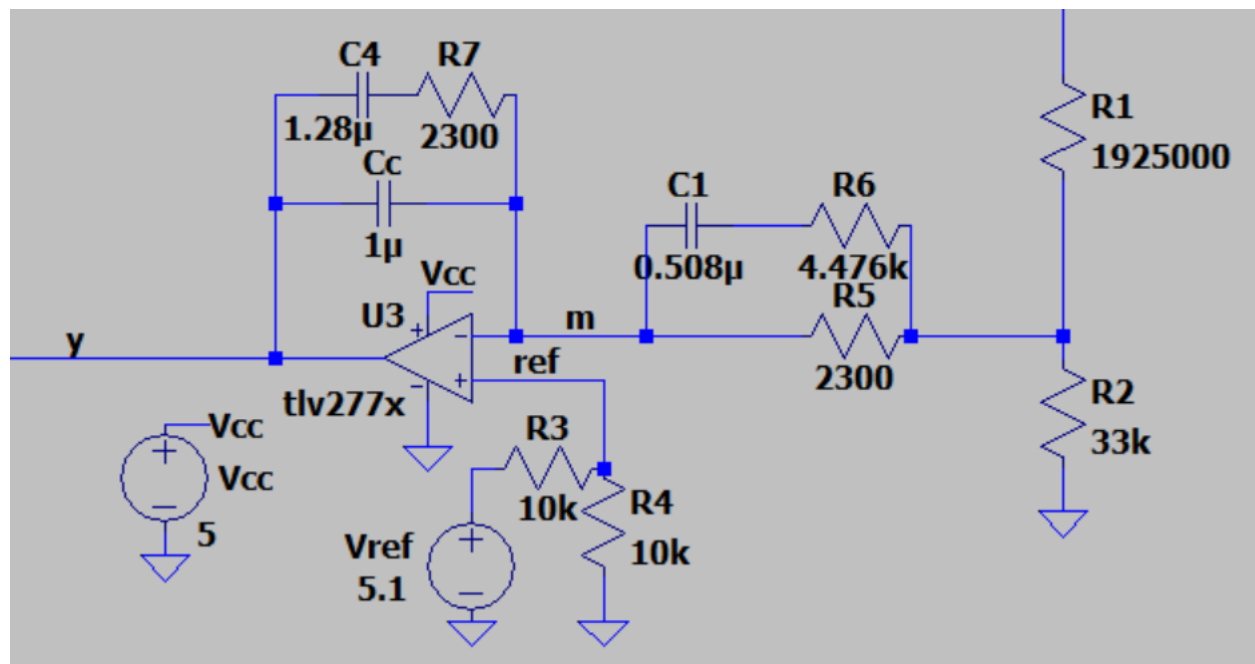
We also performed a very crude **Load Modulation test** on our compensator. We tested for 40W to be dissipated at the output with 500Ω load. We also tested for 90W by adding another 500Ω load in parallel while the converter was already running. Following a brief transient, the converter landed on 150V within spec, at roughly the same expected efficiencies just all brought down slightly as more input current was required for the same input voltages for higher output currents at 150V. No data is shown for this test.

It is also worth noting that we periodically measured our switching frequency as we measured the data above just to ensure that it was not changing too drastically and in our random measurements we never saw it leave the range of 86.2kHz to 88.03kHz.

g. Regulation Improvement

To begin, we will mention that our regulator did not behave well at higher output voltages than 100V at first. We were stumped as to why and then by simply adding twisted pairs, we improved our resilience several fold and were able to actually achieve any output voltage within rating of duty cycle and output components. The twisted pair on the control output of the regulator reduced the noise that was evidently preventing the 3525 IC from accurately setting the duty cycle.

Now, we will implement a redesign via simulation - only because our converter works really well in practical purposes as is. Our objective is to get the crossover frequency as high as possible while keeping the phase margin at least 70 degrees. This will be done through a PID compensator that allows us to pick a higher cross-over frequency that normally would have lower phase margin with just the integral compensator but will allow us to also boost the phase at that cross-over frequency. The simulation schematic below is an implementation of the full PID compensator. We couldn't quite manage to adjust the frequencies well enough to increase the phase margin in simulation, but the circuit should work in theory.



III. Appendix

- a. Only showing the inductor losses and not the more realistic lossy solution for Part 2.
Part 3 – Analytical Solution

```
clear all
clc

syms IL1 il1 VI vi VC1 vc1 IL2 il2 VC2 vc2 D d Dp
syms Ron1 RL1 VD1 Ron2 RL2 VD2 RL
VI_t = VI + vi
```

$$VI_t = VI + vi$$

$$IL1_t = IL1 + il1$$

$$IL1_t = IL_1 + il_1$$

$$VC1_t = VC1 + vc1$$

$$VC1_t = VC_1 + vc_1$$

$$IL2_t = IL2 + il2$$

$$IL2_t = IL_2 + il_2$$

$$VC2_t = VC2 + vc2$$

$$VC2_t = VC_2 + vc_2$$

$$dt = D + d$$

$$dt = D + d$$

$$dpt = 1 - dt$$

$$dpt = 1 - d - D$$

$$A = VI_t - IL1_t * (RL1 + Ron1)$$

$$A = VI + vi - (RL_1 + Ron_1) (IL_1 + il_1)$$

$$B = VC1_t - IL2_t * (RL2 + Ron2)$$

$$B = VC_1 + vc_1 - (RL_2 + Ron_2) (IL_2 + il_2)$$

$$C = -IL2_t$$

$$C = -IL_2 - il_2$$

$$Dd = -VC2t/RL$$

$$Dd =$$

$$- \frac{VC_2 + vc_2}{RL}$$

$$E = VI t - IL1 t * RL1 - VD1 - VC1 t$$

$$E = VI - VD_1 - VC_1 - vc_1 + vi - RL_1 (IL_1 + il_1)$$

$$F = VC1 t - IL2 t * RL2 - VD2 - VC2 t$$

$$F = VC_1 - VC_2 - VD_2 + vc_1 - vc_2 - RL_2 (IL_2 + il_2)$$

$$G = IL1 t - IL2 t$$

$$G = IL_1 - IL_2 + il_1 - il_2$$

$$H = IL2 t - VC2 t / RL$$

$$H =$$

$$IL_2 + il_2 - \frac{VC_2 + vc_2}{RL}$$

$$\text{eqnL1} = \text{simplify}(\text{expand}(\text{dt}*A + \text{dpt}*E))$$

$$\text{eqnL1} =$$

$$VI - VD_1 - VC_1 - vc_1 + vi + VC_1 d + VD_1 d + D vc_1 - RL_1 il_1 + d vc_1 + D VC_1 + D VD_1 - IL_1 RL_1 - D IL_1 Ron_1 - IL_1 Ron_1 d - D Ron_1 il_1 - Ron_1 d il_1$$

$$\text{eqnL2} = \text{simplify}(\text{expand}(\text{dt}*B + \text{dpt}*F))$$

$$\text{eqnL2} =$$

$$VC_1 - VC_2 - VD_2 + vc_1 - vc_2 + VC_2 d + VD_2 d + D vc_2 - RL_2 il_2 + d vc_2 + D VC_2 + D VD_2 - IL_2 RL_2 - D IL_2 Ron_2 - IL_2 Ron_2 d - D Ron_2 il_2 - Ron_2 d il_2$$

$$\text{eqnC1} = \text{simplify}(\text{expand}(\text{dt}*C + \text{dpt}*G))$$

$$\text{eqnC1} = IL_1 - IL_2 + il_1 - il_2 - IL_1 d - D il_1 - d il_1 - D IL_1$$

$$\text{eqnC2} = \text{simplify}(\text{expand}(\text{dt}*Dd + \text{dpt}*H))$$

$$\text{eqnC2} =$$

$$IL_2 + il_2 - IL_2 d - D il_2 - d il_2 - \frac{VC_2}{RL} - \frac{vc_2}{RL} - D IL_2$$

% now break it up into DC terms and first order AC terms

$$\text{eqnL1dc} = 0 == VI - VD_1 - VC_1 + D*VC_1 + D*VD_1 - IL_1*RL_1 - D*IL_1*Ron_1$$

$$\text{eqnL1dc} = 0 = V_I - V_{D1} - V_{C1} + D V_{C1} + D V_{D1} - I_{L1} R_{L1} - D I_{L1} R_{on1}$$

$$\text{eqnL1ac} = -v_{c1} + v_i + V_{C1} \cdot d + V_{D1} \cdot d + D \cdot v_{c1} - R_{L1} \cdot i_{l1} - I_{L1} \cdot R_{on1} \cdot d - D \cdot R_{on1} \cdot i_{l1}$$

$$\text{eqnL1ac} = v_i - v_{c1} + V_{C1} d + V_{D1} d + D v_{c1} - R_{L1} i_{l1} - I_{L1} R_{on1} d - D R_{on1} i_{l1}$$

$$\text{eqnL2dc} = 0 == V_{C1} - V_{C2} - V_{D2} + D \cdot V_{C2} + D \cdot V_{D2} - I_{L2} \cdot R_{L2} - D \cdot I_{L2} \cdot R_{on2}$$

$$\text{eqnL2dc} = 0 = V_{C1} - V_{C2} - V_{D2} + D V_{C2} + D V_{D2} - I_{L2} R_{L2} - D I_{L2} R_{on2}$$

$$\text{eqnL2ac} = v_{c1} - v_{c2} + V_{C2} \cdot d + V_{D2} \cdot d + D \cdot v_{c2} - R_{L2} \cdot i_{l2} - I_{L2} \cdot R_{on2} \cdot d - D \cdot R_{on2} \cdot i_{l2}$$

$$\text{eqnL2ac} = v_{c1} - v_{c2} + V_{C2} d + V_{D2} d + D v_{c2} - R_{L2} i_{l2} - I_{L2} R_{on2} d - D R_{on2} i_{l2}$$

$$\text{eqnC1dc} = 0 == I_{L1} - I_{L2} - D \cdot I_{L1}$$

$$\text{eqnC1dc} = 0 = I_{L1} - I_{L2} - D I_{L1}$$

$$\text{eqnC1ac} = i_{l1} - i_{l2} - I_{L1} \cdot d - D \cdot i_{l1}$$

$$\text{eqnC1ac} = i_{l1} - i_{l2} - I_{L1} d - D i_{l1}$$

$$\text{eqnC2dc} = 0 == I_{L2} - V_{C2}/R_L - D \cdot I_{L2}$$

$$\text{eqnC2dc} =$$

$$0 = I_{L2} - \frac{V_{C2}}{R_L} - D I_{L2}$$

$$\text{eqnC2ac} = i_{l2} - I_{L2} \cdot d - D \cdot i_{l2} - v_{c2}/R_L$$

$$\text{eqnC2ac} =$$

$$i_{l2} - I_{L2} d - D i_{l2} - \frac{v_{c2}}{R_L}$$

$$\text{eqnL1dc} = \text{subs}(\text{eqnL1dc}, [V_I \ R_{on1} \ R_{L1} \ V_{D1} \ R_{on2} \ R_{L2} \ V_{D2} \ R_L], [12 \ 0.0139 \ 0.011361 \ 0.8 \ 0.53 \ 0.02609 \ 0.7 \ 500])$$

$$\text{eqnL1dc} = 0 = 0.8000 D - 0.0114 I_{L1} - V_{C1} - 0.0139 D I_{L1} + D V_{C1} + 11.2000$$

$$\text{eqnC1dc} = \text{subs}(\text{eqnC1dc}, [V_I \ R_{on1} \ R_{L1} \ V_{D1} \ R_{on2} \ R_{L2} \ V_{D2} \ R_L], [12 \ 0.0139 \ 0.011361 \ 0.8 \ 0.53 \ 0.02609 \ 0.7 \ 500])$$

$$\text{eqnC1dc} = 0 = I_{L1} - I_{L2} - D I_{L1}$$

```
eqnL2dc = subs(eqnL2dc, [VI Ron1 RL1 VD1 Ron2 RL2 VD2 RL], [12 0.0139 0.011361
0.8 0.53 0.02609 0.7 500])
```

$$\text{eqnL2dc} = 0 = 0.7000 D - 0.0261 IL_2 + VC_1 - VC_2 - 0.5300 D IL_2 + D VC_2 - 0.7000$$

```
eqnC2dc = subs(eqnC2dc, [VI Ron1 RL1 VD1 Ron2 RL2 VD2 RL], [12 0.0139 0.011361
0.8 0.53 0.02609 0.7 500])
```

$$\text{eqnC2dc} = 0 = IL_2 - 0.0020 VC_2 - D IL_2$$

```
DCsol = solve([eqnL1dc, eqnL2dc, eqnC1dc, eqnC2dc],[IL1, IL2, VC1, VC2])
```

DCsol = *struct with fields:*

$$IL1: (1.9816e+21*D - 6.3050e+20*D^2 + 9.4576e+21)/(2.7012e+24*D^2 - 1.8010e+24*D - 1.8010e+24*D^3 + 4.5036e+23*D^4 + 4.5039e+23)$$

$$IL2: -(9.0072e+19*(D - 1)*(22*D - 7*D^2 + 105))/(2.7012e+24*D^2 - 1.8010e+24*D - 1.8010e+24*D^3 + 4.5036e+23*D^4 + 4.5039e+23)$$

$$VC1: -(1*(1.4767e+25*D - 1.4046e+25*D^2 + 3.9635e+24*D^3 + 3.6029e+23*D^4 - 5.0443e+24))/(2.7012e+24*D^2 - 1.8010e+24*D - 1.8010e+24*D^3 + 4.5036e+23*D^4 + 4.5039e+23)$$

$$VC2: (4.5036e+22*(D - 1)^2*(22*D - 7*D^2 + 105))/(2.7012e+24*D^2 - 1.8010e+24*D - 1.8010e+24*D^3 + 4.5036e+23*D^4 + 4.5039e+23)$$

```
IL1dc = DCsol.IL1
```

IL1dc =

$$\frac{9.0072e+19 (-7 D^2 + 22 D + 105)}{4.5036e+23 D^4 - 1.8010e+24 D^3 + 2.7012e+24 D^2 - 1.8010e+24 D + 4.5039e+23}$$

```
IL2dc = DCsol.IL2
```

IL2dc =

$$-\frac{9.0072e+19 (D - 1) (-7 D^2 + 22 D + 105)}{4.5036e+23 D^4 - 1.8010e+24 D^3 + 2.7012e+24 D^2 - 1.8010e+24 D + 4.5039e+23}$$

```
VC1dc = DCsol.VC1
```

VC1dc =

$$-\frac{0.1000 (3.6029e+24 D^4 + 3.9635e+25 D^3 - 1.4046e+26 D^2 + 1.4767e+26 D - 5.0443e+25)}{4.5036e+23 D^4 - 1.8010e+24 D^3 + 2.7012e+24 D^2 - 1.8010e+24 D + 4.5039e+23}$$

```
VC2dc = DCsol.VC2
```

VC2dc =

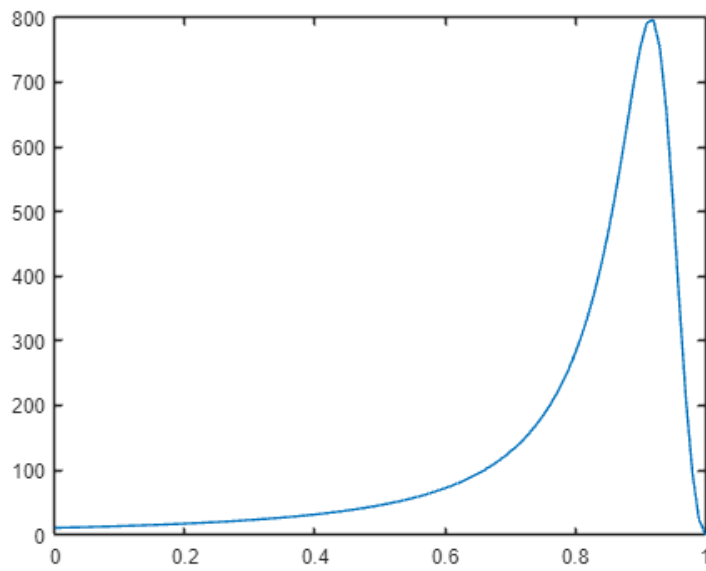
$$\frac{4.5036e+22 (D - 1)^2 (-7 D^2 + 22 D + 105)}{4.5036e+23 D^4 - 1.8010e+24 D^3 + 2.7012e+24 D^2 - 1.8010e+24 D + 4.5039e+23}$$

```
darr = 0:0.01:1
```

```
darr = 1×101
```

```
0    0.0100    0.0200    0.0300    0.0400    0.0500    0.0600 ...
```

```
plot(darr, subs(VC2dc, D, darr))
```



```
Ddc = solve(VC2dc == 150, D)
```

```
Ddc =
```

```

$$\begin{pmatrix} 0.7229 \\ 0.9747 \\ 1.0256 \\ 1.2810 \end{pmatrix}$$

```

```
Ddc = Ddc(1)
```

```
Ddc =
```

```
0.7229
```

```
IL1dc = subs(IL1dc, D, Ddc)
```

```
IL1dc =
```

```
3.9069
```

```
IL2dc = subs(IL2dc, D, Ddc)
```

```
IL2dc =
```

```
1.0826
```

```
VC1dc = subs(VC1dc, D, Ddc)
```

VC1dc =

42.2029

```
VC2dc = subs(VC2dc, D, Ddc)
```

VC2dc =

150.0000

```
eqnL1ac = subs(eqnL1ac, [VI Ron1 RL1 VD1 Ron2 RL2 VD2 RL IL1 IL2 VC1 VC2 D],  
[12 0 0.011361 0 0 0.02609 0 500 IL1dc IL2dc VC1dc VC2dc Ddc])
```

eqnL1ac = $42.2029 d - 0.0114 i_{l1} - 0.2771 v_{c1} + v_i$

```
eqnL2ac = subs(eqnL2ac, [VI Ron1 RL1 VD1 Ron2 RL2 VD2 RL IL1 IL2 VC1 VC2 D],  
[12 0 0.011361 0 0 0.02609 0 500 IL1dc IL2dc VC1dc VC2dc Ddc])
```

eqnL2ac = $150.0000 d - 0.0261 i_{l2} + v_{c1} - 0.2771 v_{c2}$

```
eqnC1ac = subs(eqnC1ac, [VI Ron1 RL1 VD1 Ron2 RL2 VD2 RL IL1 IL2 VC1 VC2 D],  
[12 0 0.011361 0 0 0.02609 0 500 IL1dc IL2dc VC1dc VC2dc Ddc])
```

eqnC1ac = $0.2771 i_{l1} - 3.9069 d - i_{l2}$

```
eqnC2ac = subs(eqnC2ac, [VI Ron1 RL1 VD1 Ron2 RL2 VD2 RL IL1 IL2 VC1 VC2 D],  
[12 0 0.011361 0 0 0.02609 0 500 IL1dc IL2dc VC1dc VC2dc Ddc])
```

eqnC2ac = $0.2771 i_{l2} - 1.0826 d - 0.0020 v_{c2}$

```
C1 = 2200*10^-6
```

C1 = 0.0022

```
C2 = 33*10^-6
```

C2 = 3.3000e-05

```
L1 = 65.7*10^-6
```

L1 = 6.5700e-05

```
L2 = 395*10^-6
```

L2 = 3.9500e-04

```
syms s
```

$$ZC1 = 1/(s*C1)$$

$$ZC1 =$$

$$\frac{454.5455}{s}$$

$$ZC2 = 1/(s*C2)$$

$$ZC2 =$$

$$\frac{3.0303e+04}{s}$$

$$ZL1 = s*L1$$

$$ZL1 = 6.5700e-05 s$$

$$ZL2 = s*L2$$

$$ZL2 = 3.9500e-04 s$$

```

ZA = (1/ZC2 + 1/RL)^-1 * (1-Ddc)^2;
ZB = ((ZA + ZL2 + 0.0262)^-1 + 1/ZC1)^-1 * (1-Ddc)^2;
VC = ZB/(ZB+ZL1 + 0.00114)*42.2029*d ;
VC1 = VC*(1-Ddc)^-2;
Vo1 = simplify(expand(VC1* ZA/(ZA+ZL2 + 0.0261)));
Vo1 = simplify(expand(subs(Vo1, RL, 500)))

```

$$Vo1 =$$

$$\frac{1.1751e+79 d (3.2080e+38 s^2 + 4.0721e+40 s + 1.8911e+45)}{2.1916e+102 s^6 + 5.9385e+104 s^5 + 2.9569e+109 s^4 + 4.5369e+111 s^3 + 1.0485e+116 s^2 + 4.2214e+117 s + 4.0475e+121}$$

```

ZA = (ZL1 + 0.0114)/((1-Ddc)^2);
ZB = (1-Ddc)^2 * (1/ZC2 + 1/RL)^-1;
ZC = ZL2 + 0.0261;
ZD = ZC1;
ZE = ((ZB + ZC)^-1 + ZD^-1)^-1;
IoverVC = -(1/ZA + 1/ZE);
VCoverI = IoverVC^-1;
VC1 = VCoverI*3.9069*d;
Vo2=simplify(expand(VC1*ZB/(ZB+ZC)));
Vo2 = simplify(expand(subs(Vo2, RL, 500)))

```

$$Vo2 =$$

$$-\frac{4.8076e+50 d (1.5149e+18 s + 2.6287e+20)}{6.9620e+58 s^4 + 2.0900e+61 s^3 + 5.2903e+65 s^2 + 9.4653e+67 s + 2.1887e+71}$$

$$ZA = (ZL1 + 0.0114)/((1-Ddc)^2);$$

```
ZB = (1-Ddc)^2 * (1/ZC2 + 1/RL)^-1;
ZC = ZL2 + 0.0261;
ZD = ZC1;
Vo3 = simplify(expand(150*d*(ZB)/((1/ZA + 1/ZD)^-1 + ZB + ZC)));
Vo3 = simplify(expand(subs(Vo3, RL, 500)))
```

Vo3 =

$$\frac{3.1901e+46 d (1.9284e+21 s^2 + 3.3460e+23 s + 1.0245e+27)}{6.9620e+58 s^4 + 2.0900e+61 s^3 + 5.2903e+65 s^2 + 9.4653e+67 s + 2.1887e+71}$$

```
ZA = (ZL1 + 0.0114)/((1-Ddc)^2);
ZB = (1/ZA + 1/ZC1)^-1 + ZC;
ZB = ZB/((1-Ddc)^2);
ZD = (1/ZC2 + 1/RL)^-1;
IoverVC = -(1/ZB + 1/ZD);
VCoverI = IoverVC^-1;
Vo4 = VCoverI*1.0826*d;
Vo4 = simplify(expand(subs(Vo4, RL, 500)))
```

Vo4 =

$$-\frac{3.0095e+25 d (7.5890e+37 s^3 + 1.8183e+40 s^2 + 1.2852e+44 s + 1.7817e+46)}{6.9620e+58 s^4 + 2.0900e+61 s^3 + 5.2903e+65 s^2 + 9.4653e+67 s + 2.1887e+71}$$

```
Gvd = simplify(expand((Vo1+Vo2+Vo3+Vo4)/d))
```

Gvd =

$$\frac{1.2627e+28 (-1.3567e+147 s^9 + 3.5851e+151 s^8 - 4.8806e+153 s^7 + 5.7700e+158 s^6 + 9.4320e+160 s^5 + 2.9569e+165 s^4 + 5.4804e+167 s^3 + 5.0082e+171 s^2 + 7.0181e+173 s + 1.6695e+177)}{5.2219e+170 s^{10} + 2.9826e+173 s^9 + 1.1056e+178 s^8 + 4.9812e+180 s^7 + 8.0680e+184 s^6 + 2.6744e+187 s^5 + 2.2341e+191 s^4 + 4.7904e+193 s^3 + 1.5319e+197 s^2 + 1.6274e+199 s + 3.0319e+202}$$

```
s_j = i*2*pi*logspace(0,4,100)
```

s_j = 1×100 complex

10⁴ ×

0.0000 + 0.0006i 0.0000 + 0.0007i 0.0000 + 0.0008i 0.0000 + 0.0008i
...

```
tf = subs(Gvd, s, s_j)
```

tf =

(695.3679 - 0.5086i 695.3791 - 0.5582i 695.3926 - 0.6126i 695.4088 - 0.6724i 695.4283 - 0.7380i 695.4518 - 0.8100i 695.4802 - 0.8890i 695.5100 - 0.9700i 695.5400 - 1.0500i 695.5700 - 1.1300i 695.6000 - 1.2100i 695.6300 - 1.2900i 695.6600 - 1.3700i 695.6900 - 1.4500i 695.7200 - 1.5300i 695.7500 - 1.6100i 695.7800 - 1.6900i 695.8100 - 1.7700i 695.8400 - 1.8500i 695.8700 - 1.9300i 695.9000 - 2.0100i 695.9300 - 2.0900i 695.9600 - 2.1700i 695.9900 - 2.2500i 696.0200 - 2.3300i 696.0500 - 2.4100i 696.0800 - 2.4900i 696.1100 - 2.5700i 696.1400 - 2.6500i 696.1700 - 2.7300i 696.2000 - 2.8100i 696.2300 - 2.8900i 696.2600 - 2.9700i 696.2900 - 3.0500i 696.3200 - 3.1300i 696.3500 - 3.2100i 696.3800 - 3.2900i 696.4100 - 3.3700i 696.4400 - 3.4500i 696.4700 - 3.5300i 696.5000 - 3.6100i 696.5300 - 3.6900i 696.5600 - 3.7700i 696.5900 - 3.8500i 696.6200 - 3.9300i 696.6500 - 4.0100i 696.6800 - 4.0900i 696.7100 - 4.1700i 696.7400 - 4.2500i 696.7700 - 4.3300i 696.8000 - 4.4100i 696.8300 - 4.4900i 696.8600 - 4.5700i 696.8900 - 4.6500i 696.9200 - 4.7300i 696.9500 - 4.8100i 696.9800 - 4.8900i 697.0100 - 4.9700i 697.0400 - 5.0500i 697.0700 - 5.1300i 697.1000 - 5.2100i 697.1300 - 5.2900i 697.1600 - 5.3700i 697.1900 - 5.4500i 697.2200 - 5.5300i 697.2500 - 5.6100i 697.2800 - 5.6900i 697.3100 - 5.7700i 697.3400 - 5.8500i 697.3700 - 5.9300i 697.4000 - 6.0100i 697.4300 - 6.0900i 697.4600 - 6.1700i 697.4900 - 6.2500i 697.5200 - 6.3300i 697.5500 - 6.4100i 697.5800 - 6.4900i 697.6100 - 6.5700i 697.6400 - 6.6500i 697.6700 - 6.7300i 697.7000 - 6.8100i 697.7300 - 6.8900i 697.7600 - 6.9700i 697.7900 - 7.0500i 697.8200 - 7.1300i 697.8500 - 7.2100i 697.8800 - 7.2900i 697.9100 - 7.3700i 697.9400 - 7.4500i 697.9700 - 7.5300i 698.0000 - 7.6100i 698.0300 - 7.6900i 698.0600 - 7.7700i 698.0900 - 7.8500i 698.1200 - 7.9300i 698.1500 - 8.0100i 698.1800 - 8.0900i 698.2100 - 8.1700i 698.2400 - 8.2500i 698.2700 - 8.3300i 698.3000 - 8.4100i 698.3300 - 8.4900i 698.3600 - 8.5700i 698.3900 - 8.6500i 698.4200 - 8.7300i 698.4500 - 8.8100i 698.4800 - 8.8900i 698.5100 - 8.9700i 698.5400 - 9.0500i 698.5700 - 9.1300i 698.6000 - 9.2100i 698.6300 - 9.2900i 698.6600 - 9.3700i 698.6900 - 9.4500i 698.7200 - 9.5300i 698.7500 - 9.6100i 698.7800 - 9.6900i 698.8100 - 9.7700i 698.8400 - 9.8500i 698.8700 - 9.9300i 698.9000 - 10.0100i 698.9300 - 10.0900i 698.9600 - 10.1700i 698.9900 - 10.2500i 699.0200 - 10.3300i 699.0500 - 10.4100i 699.0800 - 10.4900i 699.1100 - 10.5700i 699.1400 - 10.6500i 699.1700 - 10.7300i 699.2000 - 10.8100i 699.2300 - 10.8900i 699.2600 - 10.9700i 699.2900 - 11.0500i 699.3200 - 11.1300i 699.3500 - 11.2100i 699.3800 - 11.2900i 699.4100 - 11.3700i 699.4400 - 11.4500i 699.4700 - 11.5300i 699.5000 - 11.6100i 699.5300 - 11.6900i 699.5600 - 11.7700i 699.5900 - 11.8500i 699.6200 - 11.9300i 699.6500 - 12.0100i 699.6800 - 12.0900i 699.7100 - 12.1700i 699.7400 - 12.2500i 699.7700 - 12.3300i 699.8000 - 12.4100i 699.8300 - 12.4900i 699.8600 - 12.5700i 699.8900 - 12.6500i 699.9200 - 12.7300i 699.9500 - 12.8100i 699.9800 - 12.8900i 700.0100 - 12.9700i 700.0400 - 13.0500i 700.0700 - 13.1300i 700.1000 - 13.2100i 700.1300 - 13.2900i 700.1600 - 13.3700i 700.1900 - 13.4500i 700.2200 - 13.5300i 700.2500 - 13.6100i 700.2800 - 13.6900i 700.3100 - 13.7700i 700.3400 - 13.8500i 700.3700 - 13.9300i 700.4000 - 14.0100i 700.4300 - 14.0900i 700.4600 - 14.1700i 700.4900 - 14.2500i 700.5200 - 14.3300i 700.5500 - 14.4100i 700.5800 - 14.4900i 700.6100 - 14.5700i 700.6400 - 14.6500i 700.6700 - 14.7300i 700.7000 - 14.8100i 700.7300 - 14.8900i 700.7600 - 14.9700i 700.7900 - 15.0500i 700.8200 - 15.1300i 700.8500 - 15.2100i 700.8800 - 15.2900i 700.9100 - 15.3700i 700.9400 - 15.4500i 700.9700 - 15.5300i 701.0000 - 15.6100i 701.0300 - 15.6900i 701.0600 - 15.7700i 701.0900 - 15.8500i 701.1200 - 15.9300i 701.1500 - 16.0100i 701.1800 - 16.0900i 701.2100 - 16.1700i 701.2400 - 16.2500i 701.2700 - 16.3300i 701.3000 - 16.4100i 701.3300 - 16.4900i 701.3600 - 16.5700i 701.3900 - 16.6500i 701.4200 - 16.7300i 701.4500 - 16.8100i 701.4800 - 16.8900i 701.5100 - 16.9700i 701.5400 - 17.0500i 701.5700 - 17.1300i 701.6000 - 17.2100i 701.6300 - 17.2900i 701.6600 - 17.3700i 701.6900 - 17.4500i 701.7200 - 17.5300i 701.7500 - 17.6100i 701.7800 - 17.6900i 701.8100 - 17.7700i 701.8400 - 17.8500i 701.8700 - 17.9300i 701.9000 - 18.0100i 701.9300 - 18.0900i 701.9600 - 18.1700i 701.9900 - 18.2500i 702.0200 - 18.3300i 702.0500 - 18.4100i 702.0800 - 18.4900i 702.1100 - 18.5700i 702.1400 - 18.6500i 702.1700 - 18.7300i 702.2000 - 18.8100i 702.2300 - 18.8900i 702.2600 - 18.9700i 702.2900 - 19.0500i 702.3200 - 19.1300i 702.3500 - 19.2100i 702.3800 - 19.2900i 702.4100 - 19.3700i 702.4400 - 19.4500i 702.4700 - 19.5300i 702.5000 - 19.6100i 702.5300 - 19.6900i 702.5600 - 19.7700i 702.5900 - 19.8500i 702.6200 - 19.9300i 702.6500 - 20.0100i 702.6800 - 20.0900i 702.7100 - 20.1700i 702.7400 - 20.2500i 702.7700 - 20.3300i 702.8000 - 20.4100i 702.8300 - 20.4900i 702.8600 - 20.5700i 702.8900 - 20.6500i 702.9200 - 20.7300i 702.9500 - 20.8100i 702.9800 - 20.8900i 703.0100 - 20.9700i 703.0400 - 21.0500i 703.0700 - 21.1300i 703.1000 - 21.2100i 703.1300 - 21.2900i 703.1600 - 21.3700i 703.1900 - 21.4500i 703.2200 - 21.5300i 703.2500 - 21.6100i 703.2800 - 21.6900i 703.3100 - 21.7700i 703.3400 - 21.8500i 703.3700 - 21.9300i 703.4000 - 22.0100i 703.4300 - 22.0900i 703.4600 - 22.1700i 703.4900 - 22.2500i 703.5200 - 22.3300i 703.5500 - 22.4100i 703.5800 - 22.4900i 703.6100 - 22.5700i 703.6400 - 22.6500i 703.6700 - 22.7300i 703.7000 - 22.8100i 703.7300 - 22.8900i 703.7600 - 22.9700i 703.7900 - 23.0500i 703.8200 - 23.1300i 703.8500 - 23.2100i 703.8800 - 23.2900i 703.9100 - 23.3700i 703.9400 - 23.4500i 703.9700 - 23.5300i 704.0000 - 23.6100i 704.0300 - 23.6900i 704.0600 - 23.7700i 704.0900 - 23.8500i 704.1200 - 23.9300i 704.1500 - 24.0100i 704.1800 - 24.0900i 704.2100 - 24.1700i 704.2400 - 24.2500i 704.2700 - 24.3300i 704.3000 - 24.4100i 704.3300 - 24.4900i 704.3600 - 24.5700i 704.3900 - 24.6500i 704.4200 - 24.7300i 704.4500 - 24.8100i 704.4800 - 24.8900i 704.5100 - 24.9700i 704.5400 - 25.0500i 704.5700 - 25.1300i 704.6000 - 25.2100i 704.6300 - 25.2900i 704.6600 - 25.3700i 704.6900 - 25.4500i 704.7200 - 25.5300i 704.7500 - 25.6100i 704.7800 - 25.6900i 704.8100 - 25.7700i 704.8400 - 25.8500i 704.8700 - 25.9300i 704.9000 - 26.0100i 704.9300 - 26.0900i 704.9600 - 26.1700i 704.9900 - 26.2500i 705.0200 - 26.3300i 705.0500 - 26.4100i 705.0800 - 26.4900i 705.1100 - 26.5700i 705.1400 - 26.6500i 705.1700 - 26.7300i 705.2000 - 26.8100i 705.2300 - 26.8900i 705.2600 - 26.9700i 705.2900 - 27.0500i 705.3200 - 27.1300i 705.3500 - 27.2100i 705.3800 - 27.2900i 705.4100 - 27.3700i 705.4400 - 27.4500i 705.4700 - 27.5300i 705.5000 - 27.6100i 705.5300 - 27.6900i 705.5600 - 27.7700i 705.5900 - 27.8500i 705.6200 - 27.9300i 705.6500 - 28.0100i 705.6800 - 28.0900i 705.7100 - 28.1700i 705.7400 - 28.2500i 705.7700 - 28.3300i 705.8000 - 28.4100i 705.8300 - 28.4900i 705.8600 - 28.5700i 705.8900 - 28.6500i 705.9200 - 28.7300i 705.9500 - 28.8100i 705.9800 - 28.8900i 706.0100 - 28.9700i 706.0400 - 29.0500i 706.0700 - 29.1300i 706.1000 - 29.2100i 706.1300 - 29.2900i 706.1600 - 29.3700i 706.1900 - 29.4500i 706.2200 - 29.5300i 706.2500 - 29.6100i 706.2800 - 29.6900i 706.3100 - 29.7700i 706.3400 - 29.8500i 706.3700 - 29.9300i 706.4000 - 30.0100i 706.4300 - 30.0900i 706.4600 - 30.1700i 706.4900 - 30.2500i 706.5200 - 30.3300i 706.5500 - 30.4100i 706.5800 - 30.4900i 706.6100 - 30.5700i 706.6400 - 30.6500i 706.6700 - 30.7300i 706.7000 - 30.8100i 706.7300 - 30.8900i 706.7600 - 30.9700i 706.7900 - 31.0500i 706.8200 - 31.1300i 706.8500 - 31.2100i 706.8800 - 31.2900i 706.9100 - 31.3700i 706.9400 - 31.4500i 706.9700 - 31.5300i 707.0000 - 31.6100i 707.0300 - 31.6900i 707.0600 - 31.7700i 707.0900 - 31.8500i 707.1200 - 31.9300i 707.1500 - 32.0100i 707.1800 - 32.0900i 707.2100 - 32.1700i 707.2400 - 32.2500i 707.2700 - 32.3300i 707.3000 - 32.4100i 707.3300 - 32.4900i 707.3600 - 32.5700i 707.3900 - 32.6500i 707.4200 - 32.7300i 707.4500 - 32.8100i 707.4800 - 32.8900i 707.5100 - 32.9700i 707.5400 - 33.0500i 707.5700 - 33.1300i 707.6000 - 33.2100i 707.6300 - 33.2900i 707.6600 - 33.3700i 707.6900 - 33.4500i 707.7200 - 33.5300i 707.7500 - 33.6100i 707.7800 - 33.6900i 707.8100 - 33.7700i 707.8400 - 33.8500i 707.8700 - 33.9300i 707.9000 - 34.0100i 707.9300 - 34.0900i 707.9600 - 34.1700i 707.9900 - 34.2500i 708.0200 - 34.3300i 708.0500 - 34.4100i 708.0800 - 34.4900i 708.1100 - 34.5700i 708.1400 - 34.6500i 708.1700 - 34.7300i 708.2000 - 34.8100i 708.2300 - 34.8900i 708.2600 - 34.9700i 708.2900 - 35.0500i 708.3200 - 35.1300i 708.3500 - 35.2100i 708.3800 - 35.2900i 708.4100 - 35.3700i 708.4400 - 35.4500i 708.4700 - 35.5300i 708.5000 - 35.6100i 708.5300 - 35.6900i 708.5600 - 35.7700i 708.5900 - 35.8500i 708.6200 - 35.9300i 708.6500 - 36.0100i 708.6800 - 36.0900i 708.7100 - 36.1700i 708.7400 - 36.2500i 708.7700 - 36.3300i 708.8000 - 36.4100i 708.8300 - 36.4900i 708.8600 - 36.5700i 708.8900 - 36.6500i 708.9200 - 36.7300i 708.9500 - 36.8100i 708.9800 - 36.8900i 709.0100 - 36.9700i 709.0400 - 37.0500i 709.0700 - 37.1300i 709.1000 - 37.2100i 709.1300 - 37.2900i 709.1600 - 37.3700i 709.1900 - 37.4500i 709.2200 - 37.5300i 709.2500 - 37.6100i 709.2800 - 37.6900i 709.3100 - 37.7700i 709.3400 - 37.8500i 709.3700 - 37.9300i 709.4000 - 38.0100i 709.4300 - 38.0900i 709.4600 - 38.1700i 709.4900 - 38.2500i 709.5200 - 38.3300i 709.5500 - 38.4100i 709.5800 - 38.4900i 709.6100 - 38.5700i 709.6400 - 38.6500i 709.6700 - 38.7300i 709.7000 - 38.8100i 709.7300 - 38.8900i 709.7600 - 38.9700i 709.7900 - 39.0500i 709.8200 - 39.1300i 709.8500 - 39.2100i 709.8800 - 39.2900i 709.9100 - 39.3700i 709.9400 - 39.4500i 709.9700 - 39.5300i 710.0000 - 39.6100i 710.0300 - 39.6900i 710.0600 - 39.7700i 710.0900 - 39.8500i 710.1200 - 39.9300i 710.1500 - 40.0100i 710.1800 - 40.0900i 710.2100 - 40.1700i 710.2400 - 40.2500i 710.2700 - 40.3300i 710.3000 - 40.4100i 710.3300 - 40.4900i 710.3600 - 40.5700i 710.3900 - 40.6500i 710.4200 - 40.7300i 710.4500 - 40.8100i 710.4800 - 40.8900i 710.5100 - 40.9700i 710.5400 - 41.0500i 710.5700 - 41.1300i 710.6000 - 41.2100i 710.6300 - 41.2900i 710.6600 - 41.3700i 710.6900 - 41.4500i 710.7200 - 41.5300i 710.7500 - 41.6100i 710.7800 - 41.6900i 710.8100 - 41.7700i 710.8400 - 41.8500i 710.8700 - 41.9300i 710.9000 - 42.0100i 710.9300 - 42.0900i 710.9600 - 42.1700i 710.9900 - 42.2500i 711.0200 - 42.3300i 711.0500 - 42.4100i 711.0800 - 42.4900i 711.1100 - 42.5700i 711.1400 - 42.6500i 711.1700 - 42.7300i 711.2000 - 42.8100i 711.2300 - 42.8900i 711.2600 - 42.9700i 711.2900 - 43.0500i 711.3200 - 43.1300i 711.3500 - 43.2100i 711.3800 - 43.2900i 711.4100 - 43.3700i 711.4400 - 43.4500i 711.4700 - 43.5300i 711.5000 - 43.6100i 711.5300 - 43.6900i 711.5600 - 43.7700i 711.5900 - 43.8500i 711.6200 - 43.9300i 711.6500 - 44.01

```
tf_mag = double(abs(tf))
```

```
tf_mag = 1×100
```

```
104 ×
```

```
0.0695    0.0695    0.0695    0.0695    0.0695    0.0695    0.0695 ...
```

```
tf_ang = unwrap(angle(double(tf)))*(180/pi)
```

```
tf_ang = 1×100
```

```
-0.0419    -0.0460    -0.0505    -0.0554    -0.0608    -0.0667    -0.0732 ...
```

```
semilogx(logspace(0,4,100), 20*log10(tf_mag))  
ylabel('Magnitude (dB)')  
title('Analytical Solution Frequency Response')  
hold on  
yyaxis right  
semilogx(logspace(0,4,100), tf_ang)  
grid on  
ylabel('Phase (degrees)')  
xlabel('Frequency (Hz)')
```

